

Customer Shopping Behavior Analysis Using Sql:

use customer_behavior;

select * from customer;

-- Q1. What is the total revenue generated by male vs. female customers?

select gender, **sum**(purchase_amount) **as** total_revenue **from** customer **group by** gender;

	gender	total_revenue
▶	Male	157890
	Female	75191

-- Q2. Which customers used a discount but still spent more than the average purchase amount?

select customer_id, purchase_amount **from** customer **where** discount_applied='Yes' and purchase_amount > (**select avg**(purchase_amount) **from** customer);

	customer_id	purchase_amount
▶	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72
	16	81
	20	90
	22	62
	24	88
	29	94
	32	79
	33	67

-- Q3. Which are the top 5 products with the highest average review rating?

```
select item_purchased,round(avg(review_rating),2) as avg_rating from customer group by
item_purchased order by avg_rating desc limit 5;
```

	item_purchased	avg_rating
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Handbag	3.78

-- Q4. Compare the average Purchase Amounts between Standard and Express Shipping.

```
select shipping_type,round(avg(purchase_amount),2) as avg_amount_spend from
customer where shipping_type in ('standard','express') group by shipping_type;
```

Result Grid			Filter Rows:
	shipping_type	avg_amount_spend	
▶	Express	60.48	
	Standard	58.46	

-- Q5. Do subscribed customers spend more? Compare average spend and total revenue

-- between subscribers and non-subscribers.

```
select subscription_status,count(customer_id) as
total_customer,round(avg(purchase_amount),2) as avg_spend, sum(purchase_amount) as
total_revenue from customer group by subscription_status;
```

	subscription_status	total_customer	avg_spend	total_revenue
▶	Yes	1053	59.49	62645
	No	2847	59.87	170436

-- Q6. Which 5 products have the highest percentage of purchases with discounts applied?

```
select item_purchased, round((count(*)*100/(select count(*) from customer c2 where
c2.item_purchased=c1.item_purchased)),2) as discount_rate from customer c1 where
discount_applied='Yes' group by item_purchased order by discount_rate desc limit 5;
```

	item_purchased	discount_rate
▶	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

-- Q7. Segment customers into New, Returning, and Loyal based on their total

-- number of previous purchases, and show the count of each segment.

```
select if(previous_purchases=1,'New',if(previous_purchases<=10,'returning','loyal')) as
customer_segmentation,count(*) from customer group by customer_segmentation;
```

```
select
case
when previous_purchases=1 then 'new'
when previous_purchases<=10 then 'returning'
else 'loyal'
end as customer_segmentation, count(*) as no_of_customer from customer group by
customer_segmentation;
```

	customer_segmentation	no_of_customer
▶	loyal	3116
	returning	701
	new	83

-- Q8. What are the top 3 most purchased products within each category?

```
with count_item as (  
    select category,item_purchased,count(customer_id) as total_count,  
           row_number() over (partition by category order by count(customer_id) desc) as  
item_rank from customer  
    group by category,item_purchased)
```

```
select item_rank,category,item_purchased,total_count from count_item where  
item_rank<=3;
```

	item_rank	category	item_purchased	total_count
▶	1	Accessories	Jewelry	171
	2	Accessories	Sunglasses	161
	3	Accessories	Belt	161
	1	Clothing	Blouse	171
	2	Clothing	Pants	171
	3	Clothing	Shirt	169
	1	Footwear	Sandals	160
	2	Footwear	Shoes	150
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	161

-- Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

```
select subscription_status, count(customer_id) as repeat_buyer from customer where  
previous_purchases>5  
group by subscription_status;
```

	subscription_status	repeat_buyer
▶	Yes	958
	No	2518

-- Q10. What is the revenue contribution of each age group?

```
select age_group, sum(purchase_amount) as total_revenue from customer group by  
age_group order by total_revenue desc;
```

	age_group	total_revenue
▶	Young Adult	62143
	Middle-aged	59197
	Adult	55978
	Senior	55763